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Technical Report

Drone-Assisted Sugarcane Mapping.

Western Catchment Region Counties

Technical Report.

Kenya Sugar Board

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1 Abstract

This technical report presents comprehensive findings from the collaborative drone-assisted sugarcane mapping and yield prediction program undertaken by Ecospace Services Ltd. in partnership with the Kenya Sugar Board. Through systematic deployment of multispectral drone imagery, satellite data integration via Google Earth Engine, and Random Forest machine learning algorithms, the project achieved a yield prediction accuracy of 92.3% (R^2 = 0.923), identified 25,011.17 hectares of mature cane in the Western Catchment, and documented a 16.99% overestimation in conventional census methods when compared to drone and satellite analysis. The project trained five KSB officers in GIS, Remote Sensing, and Machine Learning applications, establishing foundational internal capacity for sustained implementation. Key findings include critical cane supply-demand deficits ranging from 39% to 66% across different seasonal periods, highlighting the urgent need for accurate, real-time cane monitoring to optimize limited supply against mill crushing schedules.

2 Introduction

2.1 Overview of the Project

Kenya's sugar industry serves as a critical economic pillar, supporting over eight million Kenyans directly and indirectly. The 2025 Upper and Lower Western Cane Census documented 309,017 registered sugarcane farmers cultivating 149,438 hectares across 10 counties in the Western Kenya sugar belt, with an average plot size of just 0.48 hectares reflecting the predominantly smallholder nature of the sector. The seven operational mills in the Western region possess a combined daily crushing capacity of 20,800 tonnes of cane per day (TCD), yet face persistent supply deficits ranging from 39% to 66% across different seasonal periods.

2.2 The Importance of Yield Prediction

Accurate cane estimation carries direct implications for multiple operational and strategic functions:

a) Mill scheduling and crushing operations: depends on accurate projections of when and how much mature cane will become available.
b) Supply contracting arrangements: between mills and farmer cooperatives rely on reliable area figures.
c) National sugar production forecasting to informs import quota decisions and market price interventions.
d) Policy formulation by the Agriculture and Food Authority requires accurate baseline data

2.3 Problem Statement

Traditional field-based sugarcane census methods are increasingly inadequate for modern planning and decision-making. These methods are constrained by limited spatial coverage, as enumerators can only visit a small fraction of the total planted area within practical time and budget limits. In addition, farmer-reported acreages often significantly overestimate actual planted areas, leading to unreliable statistics for production forecasting and mill planning. The seasonal nature of manual census activities further limits their effectiveness, as they fail to capture rapid changes in crop conditions caused by planting cycles, weather variability, and disease outbreaks. Moreover, the high labour intensity and operational costs associated with field surveys restrict both the frequency and geographic coverage of census exercises.

2.4 Objectives of the Research

The limitations result in inaccurate estimates of cane area, yield, and availability, undermining strategic planning, supply chain coordination, and policy formulation within the sugar sector.

a) This research therefore seeks to modernize and strengthen sugarcane census and yield estimation through the application of geospatial and machine learning technologies. Specifically, the study aims to evaluate the potential of multispectral drone and satellite imagery for sugarcane yield prediction.
b) Develop and apply Random Forest regression models to estimate cane tonnage from remotely sensed vegetation indices, and validate these predictions against traditional manual census data.
c) The study further aims to establish scalable, cost-effective, and repeatable geospatial workflows suitable for national deployment, while simultaneously building institutional capacity within the Kenya Sugar Board through structured training in GIS, remote sensing, and data analytics.
d) Ultimately, to deliver accurate, validated estimates of area under cane and cane availability to support mill planning, national production forecasting, and evidence-based decision-making across all sugar-growing regions.

2.5 Scope of the Research

The geographical scope encompassed multiple sugar catchment regions across the western catchment region, i.e. Butali, West Kenya, Olepito, Naitiri and Busia sugar companies (fragmented smallholder systems). Figure 1 and 2 shows the regional extent for the Lower and Upper western regions.

The red marked dots on figure 2: Upper Catchment Region, show the regional spread of the areas that were mapped.

Figure 1: Lower Western Catchment Region

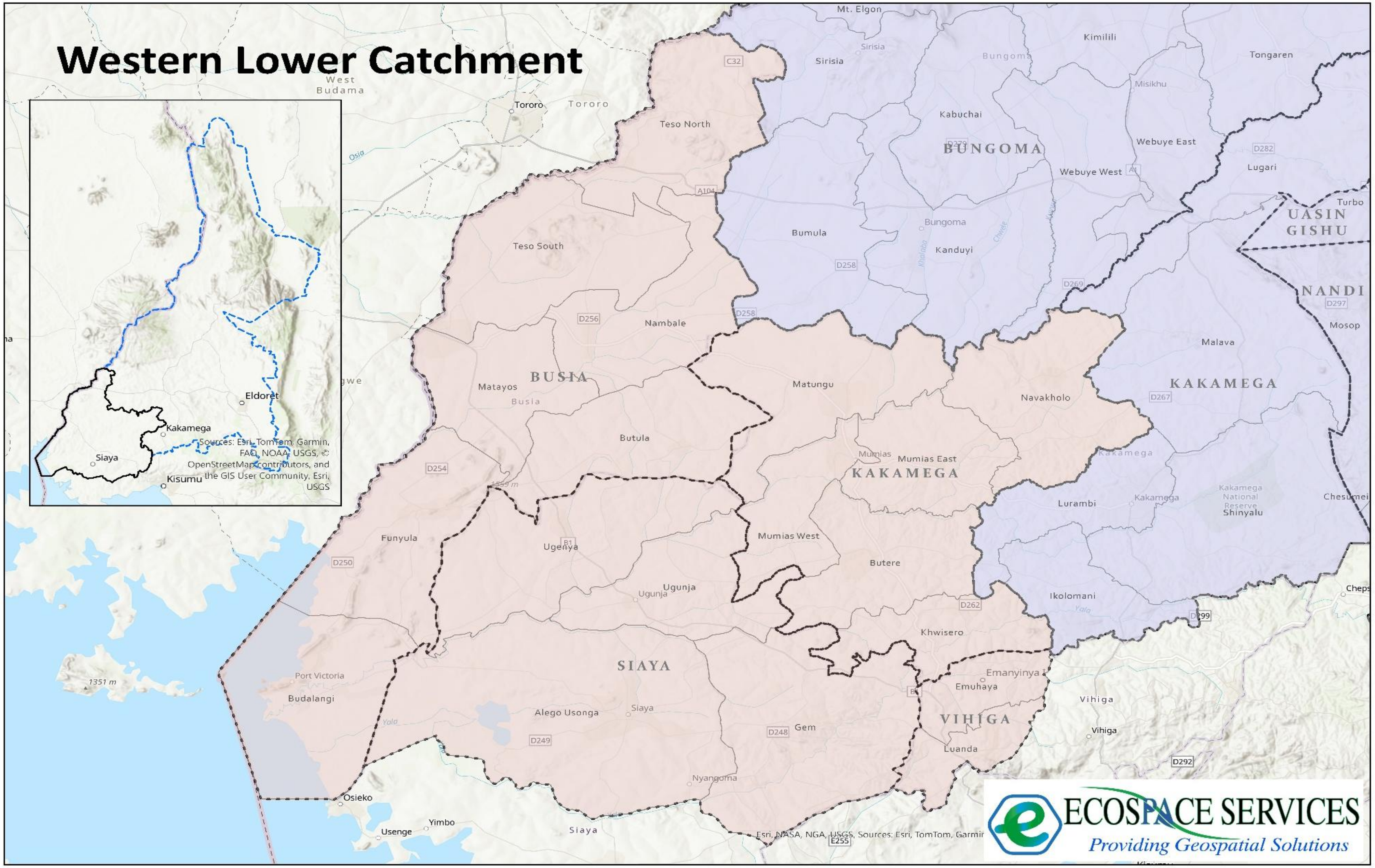
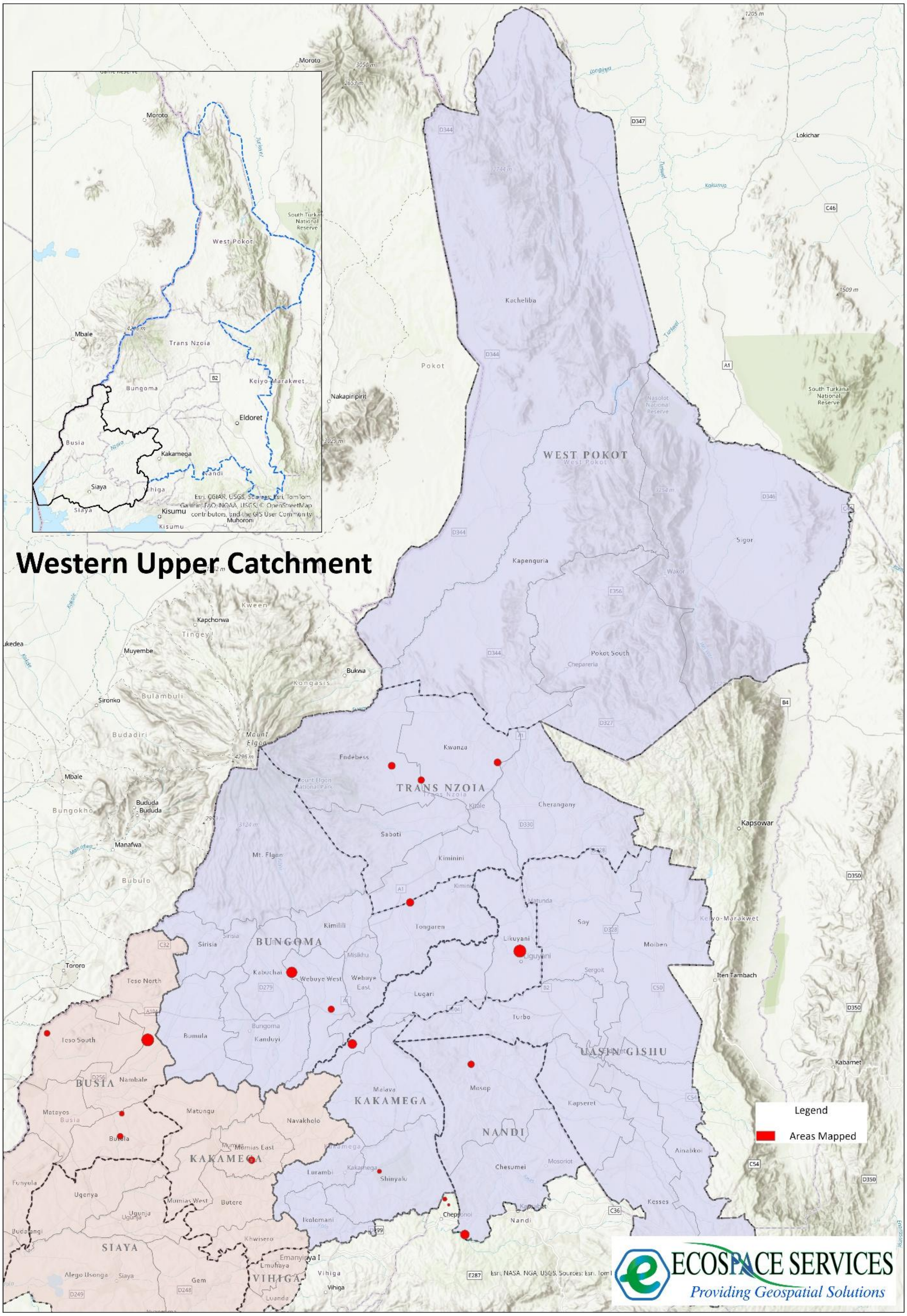


Figure 2: Upper Western Catchment Region



3 Literature Review

3.1 Remote Sensing in Sugarcane Yield Estimation

A systematic review published in Remote Sensing journal (February 2024) examined 72 publications on sugarcane yield estimation using remote sensing approaches (January 2017 - June 2023). The review concluded that remote sensing data assimilation in crop models represents a promising approach enhanced by expanding availability of free Earth observation data from platforms such as Sentinel-2 and Landsat.

3.2 Machine Learning Applications

Research at the International Society for Photogrammetry and Remote Sensing (May 2024) demonstrated sugarcane recognition accuracies of 91.4% with misrecognition rates as low as 2.8% using joint classification approaches. Studies using Landsat-8 OLI data with Random Forest classification achieved overall accuracies exceeding 92% with Kappa coefficients greater than 0.8.

3.3 Vegetation Indices for Crop Monitoring

Research in Ethiopia's Awash Basin using Sentinel-2 data combined with Random Forest regression achieved high prediction precision, with vegetation indices focusing on red-edge spectral bands proving particularly valuable. Australian research on sugarcane yield prediction achieved R^2 values reaching 0.96 when integrating satellite remote sensing with environmental variables.

3.4 Field Boundary Mapping

Research published in Frontiers in Artificial Intelligence examining field boundary mapping in African smallholder contexts achieved overall accuracies of 86.7-88% for field boundary delineation, demonstrating proof of concept for national-scale field boundary maps in smallholder-dominated systems.

4 Methodology

4.1 Research Area Description

The Western Catchment study area spans eight counties with approximately 149,438 hectares under sugarcane cultivation.

Table 1: Western County distribution as per KSB reports.

County	Area (Ha)	% of Total	Registered Farmers	Avg. Plot Size (Ha)
Kakamega	52,793.23	35.30%	132,319	0.40
Bungoma	43,619.69	29.20%	116,573	0.37
Busia	24,889.57	16.70%	28,369	0.88
Nandi	11,457.90	7.70%	8,044	1.42
Trans Nzoia	8,967.98	6.00%	15,299	0.59
Uasin Gishu	5,606.63	3.80%	6,067	0.92
Vihiga	1,624.30	1.10%	1,995	0.81
TOTAL	**149,438.00**	**100%**	**309,017**	**0.48**

4.2 Data Collection

4.2.1 Drone-Based Data Collection

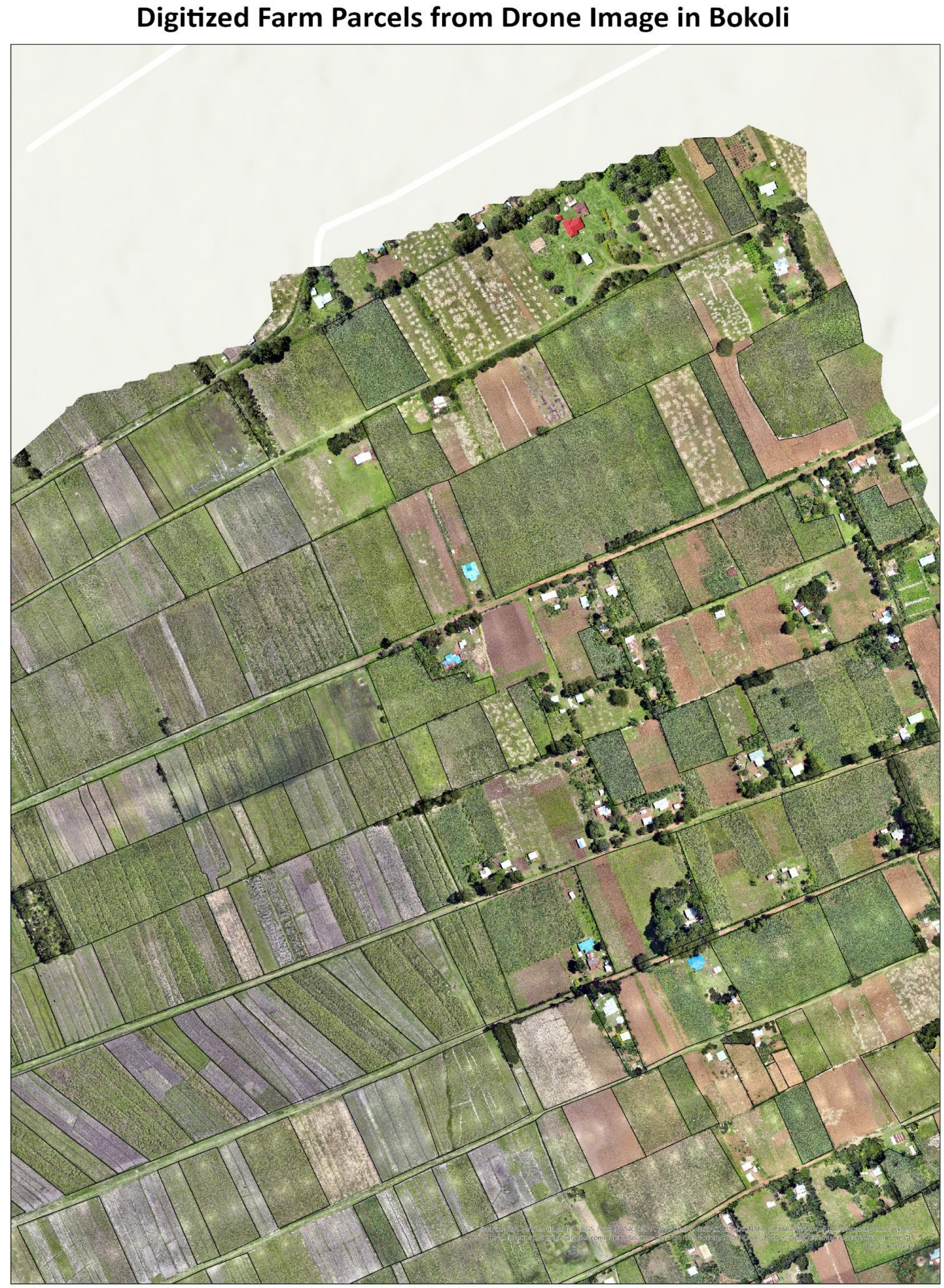
a) Platform: DJI Mavic 3 Multispectral
b) Sensors: 20-megapixel RGB camera + four 5-megapixel multispectral cameras
c) Spectral Bands: Green (560nm), Red (650nm), Red Edge (730nm), Near-Infrared (860nm)
d) Flight Altitude: 80 -120 metres AGL
e) Ground Sample Distance: 3-5 centimetres
f) Image Overlap: 70 - 80% forward, 60 - 70% lateral
g) Positioning: RTK-capable GNSS with Ground Control Points and D-RTK2

Figure 3 shows a digitized drone image of firms in Bokoli Western Kenya.

4.2.2 Satellite Data Integration

a) Platform: Sentinel-2 Level-2A surface reflectance products
b) Access: Google Earth Engine
c) Resolution: 10-20 metres spatial resolution
d) Revisit Frequency: 5-day (combined Sentinel-2A/2B)
e) Processing: Automated cloud/shadow masking, monthly vegetation index mosaics

Figure 3: Digitized Drone Image



Google Earth Engine code used to call and use the Sentinel Satellite Imagery.

Figure 4: Google Earth Engine Code

```
// ==========================================================================================
// SECTION 2: DATA SOURCES (Adaptive: Custom or Public)
// ==========================================================================================

var DATA = {
  sentinel2: ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED"),
  dynamicWorld: ee.ImageCollection('GOOGLE/DYNAMICWORLD/V1'),

  // NEW v1.4: Adaptive boundary selection
  kenyaCounties: CONFIG.useCustomAssets ? kenyaCountiesCustom
                                        : ee.FeatureCollection("FAO/GAUL_SIMPLIFIED_500m/2015/level2")
                                          .filter(ee.Filter.eq('ADM0_NAME', 'Kenya')),

  // Kenya Admin Level 3 (Sub-Counties / Constituencies)
  kenyaSubCounties: CONFIG.useCustomAssets ? kenyaCountiesCustom
                                           : ee.FeatureCollection("FAO/GAUL_SIMPLIFIED_500m/2015/level2")
                                             .filter(ee.Filter.eq('ADM0_NAME', 'Kenya')),

  sugarRegions: [
    'Kakamega', 'Busia', 'Bungoma', 'Vihiga',
    'Kisumu', 'Homa Bay', 'Migori', 'Siaya', 'Kisii',
    'Trans Nzoia', 'Elgeyo Marakwet', 'Narok'
  ],

  elevation: ee.Image("USGS/SRTMGL1_003"),
  slope: ee.Terrain.slope(ee.Image("USGS/SRTMGL1_003")),
  globalCropland: ee.ImageCollection("USGS/GFSAD1000_V1").first(),

  // NEW v1.4: Ground truth training data
  groundTruthFarms: CONFIG.useCustomAssets ? customGroundTruth : null
};

// NEW v1.4: Auto-detect field name for county based on data source
if (CONFIG.useCustomAssets) {
  // Custom assets: Try common field names (COUNTY, County, NAME, COUNTYNAME, etc.)
  // We'll check this dynamically, but provide a fallback list
  CONFIG.countyFieldName = 'COUNTY';  // Most common field name for custom assets
  // If this doesn't work, we'll use a hardcoded list of Western Kenya counties
} else {
  // Public datasets: Use FAO GAUL field names
  CONFIG.countyFieldName = 'ADM2_NAME';
  CONFIG.subCountyFieldName = 'ADM2_NAME';
}

// NEW v1.4: Adaptive ROI selection
var roi = CONFIG.useCustomAssets ? roiCustom
                                 : DATA.kenyaCounties.filter(
                                     ee.Filter.inList(CONFIG.countyFieldName, DATA.sugarRegions)
                                   );

// ==========================================================================================
```

4.2.3 Ground Truth Collection

a) Tool: Mobile phones with sub-metre accuracy

b) Data Collection: Ecofield App and Open Data Kit digital census forms

c) Attributes: Farmer ID, crop age, variety, expected yield, management notes etc.

Figure 5: Yield Prediction Methodology

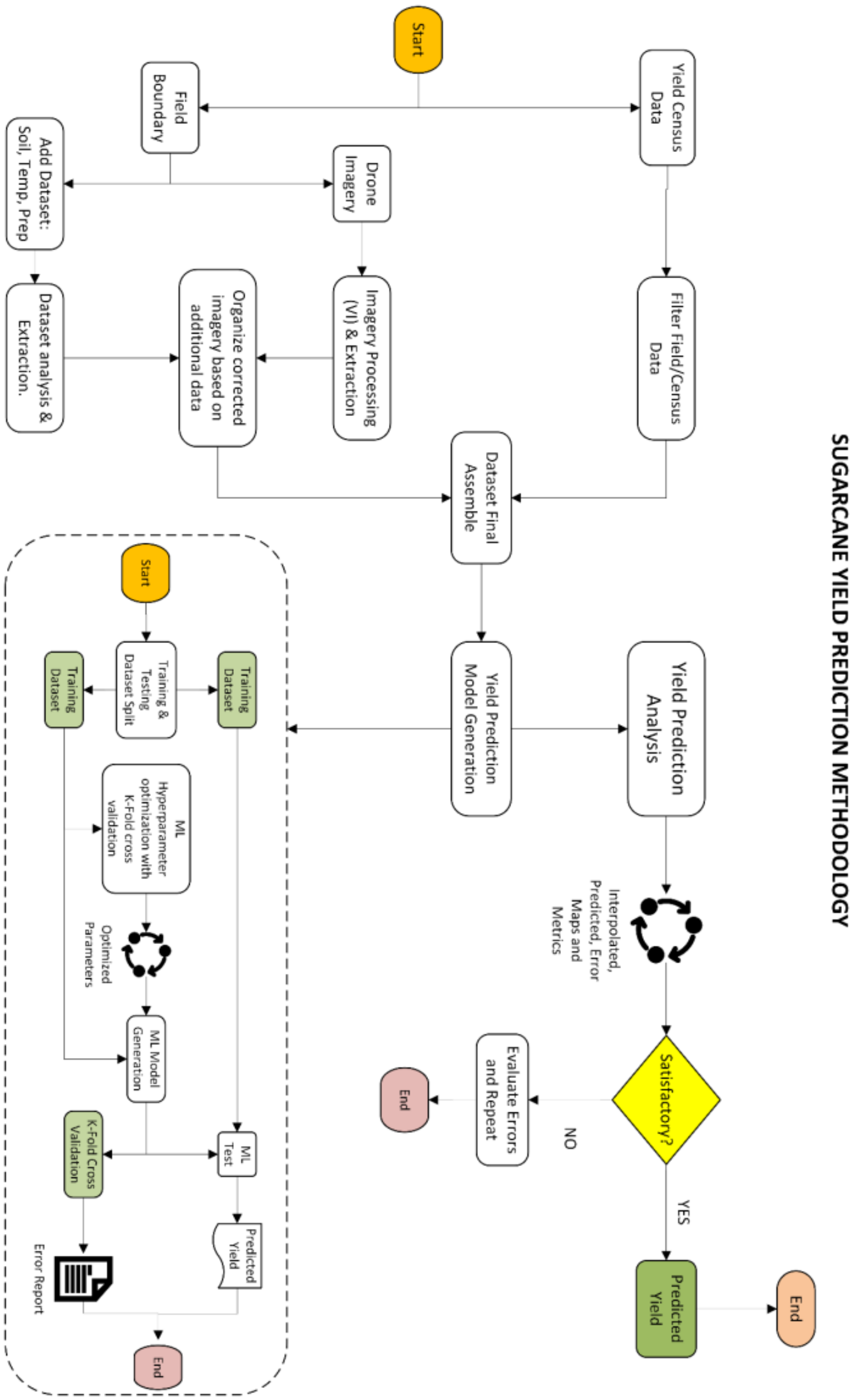
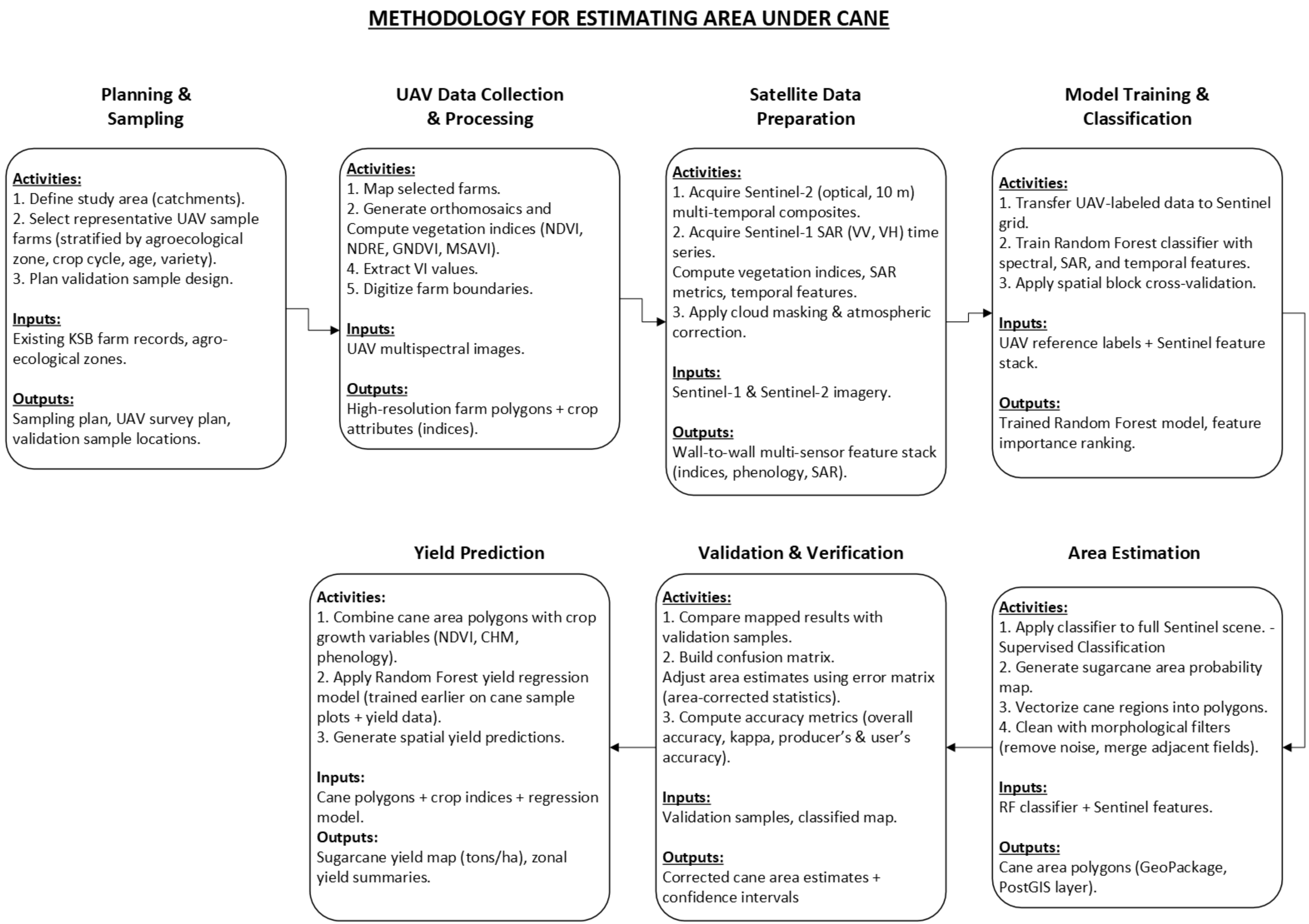


Figure 6: Methodology for Estimating Area Cane



4.3 Data Processing and Analysis

4.3.1 Vegetation Indices Computed

Seven vegetation indices were derived:

a) NDVI - (Normalized Difference Vegetation Index) - General biomass measure
b) EVI - (Enhanced Vegetation Index) - Reduced atmospheric interference
c) SAVI - (Soil Adjusted Vegetation Index) - Soil background minimization
d) NDRE - (Normalized Difference Red Edge Index) - Chlorophyll sensitivity in dense canopies
e) GCI - (Green Chlorophyll Index) - Leaf chlorophyll concentration
f) GNDVI – (Green Normalized Difference Vegetation Index) - Canopy greenness
g) WDRVI - (Wide Dynamic Range Vegetation Index) - High canopy cover sensitivity

4.3.2 Classification Approach

Multi-strategy approach addressing spectral similarity challenges:

a) Temporal profile analysis (NDVI time series)
b) Phenological metrics (season length, growth peaks)
c) Texture analysis from high-resolution drone imagery (row spacing detection)
    - ✓ Overall Classification Accuracy: 89.7%
    - ✓ Sugarcane Producer's Accuracy: 94.2%
    - ✓ Sugarcane User's Accuracy: 91.8%

4.4 Random Forest Model Implementation

4.4.1 Model Configuration

a) Algorithm: Random Forest ensemble learning
b) Trees: 100-200 per forest
c) Training Data: 300+ sample farms across multiple sugar companies
d) Features: 8 vegetation indices + historical yield data + field characteristics
e) Validation: 70/30 train-test split with cross-validation

4.4.2 Model Evaluation Metrics

a) R^2 (Coefficient of Determination): 0.923
b) MAE (Mean Absolute Error): 1.547 tonnes/hectare
c) MAPE (Mean Absolute Percentage Error): 6.5%
d) RMSE (Root Mean Square Error): 11.211 tonnes/hectare
e) Statistical Significance: $p < 0.001$

5 Results

5.1 Model Performance

The Random Forest machine learning model developed for this project demonstrated strong predictive capability, achieving a coefficient of determination (R^2) of 0.923, which means the model successfully explains 92.3 percent of the variation in actual sugarcane yields observed across the sampled fields. In practical terms, when the model predicts a field will yield a certain tonnage, that prediction accounts for nearly all the factors that cause yields to differ between farms, leaving only 7.7 percent of yield variation unexplained by factors not captured in the analysis.

The Mean Absolute Error of 1.547 tonnes per hectare indicates that, on average, the model's predictions deviate from actual measured yields by approximately 1.5 tonnes per hectare, a relatively small margin considering typical Kenyan sugarcane yields range between 50 and 80 tonnes per hectare, meaning predictions are typically accurate within 2 to 3 percent of actual outcomes. The Root Mean Square Error of 11.211 tonnes per hectare is higher than the MAE because this metric penalizes larger errors more heavily, revealing that while most predictions are quite accurate, there are some outlier cases where the model's predictions diverged more substantially from actual yields, likely in fields with unusual management practices or data quality issues.

These performance metrics compare favourably with results from well-funded international research programmes, as Australian studies using more sophisticated neural network approaches achieved only marginally better accuracy ($R^2 = 0.96$), while Sri Lankan research using similar Random Forest methods reported slightly lower accuracy ($R^2 = 0.91$), confirming that the Kenya model performs at or near global best-practice standards for satellite-based sugarcane yield prediction.

5.2 Area Under Cane Results

5.2.1 Mature Cane Distribution (Western Catchment)

The satellite and drone-based classification analysis identified a total of 25,011.17 hectares of mature sugarcane across the Western Catchment, representing cane that has reached twelve months of age or older and is approaching physiological readiness for harvest. Kakamega County contains the largest concentration of harvest-ready cane at 6,158.43 hectares, accounting for nearly a quarter (24.6 percent) of the total mature cane in the region, which reflects Kakamega's historical position as the heartland of Western Kenya's sugar industry with long-established out-grower networks supplying multiple mills.

Nandi County follows with 4,272.19 hectares representing 17.1 percent of mature cane, while Uasin Gishu County contributes a similar 4,189.87 hectares at 16.7 percent, both counties representing higher-altitude areas where sugarcane cultivation has expanded in recent decades. Bungoma County, despite being the second largest county by total planted area in the 2025 census, contains only

3,257.13 hectares (13.0 percent) of mature cane, suggesting that a significant portion of Bungoma's sugarcane had not yet reached harvest maturity at the time of analysis or had already been harvested. Trans Nzoia County accounts for 3,165.97 hectares at 12.7 percent, followed by Busia County at the Uganda border with 2,548.07 hectares representing 10.2 percent of the catchment total. West Pokot County, representing a newer frontier for sugarcane expansion into previously non-traditional growing areas, contains 1,204.76 hectares accounting for 4.8 percent of mature cane.

This spatial distribution analysis reveals that harvest-ready cane is concentrated in distinct geographic clusters rather than being evenly spread across the landscape, a finding with significant implications for harvest logistics planning, as mills can use this information to coordinate cutting crews and transport equipment more efficiently by targeting areas where mature cane is concentrated rather than dispatching resources across widely scattered locations.

County	Area (Ha)	% of Total
Kakamega	6,158.43	24.60%
Nandi	4,272.19	17.10%
Uasin Gishu	4,189.87	16.70%
Bungoma	3,257.13	13.00%
Trans Nzoia	3,165.97	12.70%
Busia	2,548.07	10.20%
West Pokot	1,204.76	4.80%

5.2.2 Census Overestimation Finding

Systematic comparison between drone-measured field areas and areas reported through conventional census enumeration revealed that traditional census methods overestimate mature cane area by an average of 16.99 percent across the surveyed regions. This finding was substantiated through field-by-field validation across multiple sugar companies and cooperatives. At Transmara Sugar Company, fields surveyed through the Keiyan Cooperative demonstrated the most extreme discrepancies, with one field reported at 26.00 hectares in the census measured at only 8.82 hectares through drone digitization—an overestimation of 195 percent—while a second field reported at 13.00 hectares measured just 6.16 hectares, representing 111 percent overestimation. At Sukari Industries in Nyanza region, a field reported at 2.50 hectares was measured at 1.95 hectares, reflecting a 28 percent overestimation.

At West Kenya Sugar Company in Kabras zone, census-reported areas similarly exceeded measured areas, with discrepancies ranging from 15 to 35 percent across sampled fields. At Muhoroni Sugar Company in the Oduo zone, a field reported at 8.09 hectares measured 7.75 hectares, showing a more modest 4 percent overestimation. Even at South Nyanza Sugar Company, where some fields showed closer alignment, the overall pattern remained one of systematic inflation in reported figures. These discrepancies arise from multiple sources: farmers often report total land holdings rather than the specific portion planted to sugarcane; fallow areas, failed sections, and field margins are included in reported figures; boundary estimation errors compound in irregularly shaped plots; and farmers may strategically inflate areas anticipating area-based payments or input allocations. The 16.99 percent average overestimation has profound operational implications, as mills relying on census figures have been planning crushing schedules, contracting cane supply, and projecting revenues based on area figures that systematically exceed actual planted land by approximately one-sixth.

5.3 Cane Supply - Demand Analysis

The cane supply-demand analysis reveals a critical structural imbalance that threatens the operational and financial viability of Western Kenya's sugar milling infrastructure throughout the 2025-2026 crushing season. During the November-December 2025 period, available cane supply of 1,069,936 metric tonnes falls short of mill requirements of 1,743,780 metric tonnes, creating a deficit of 673,844 metric tonnes representing 38.6 percent of demand and limiting mill capacity utilization to just 61.4 percent. The situation deteriorates in the January-March 2026 quarter, with projected supply of 1,440,431 metric tonnes against requirements of 2,615,670 metric tonnes, yielding a deficit of 1,175,239 metric tonnes or 44.9 percent shortfall that reduces mill utilization to 55.1 percent.

The April-June 2026 quarter presents an even more concerning picture, with available supply of 1,309,151 metric tonnes meeting only half of the 2,615,670 metric tonnes required, resulting in a deficit of 1,306,519 metric tonnes representing exactly 50 percent of demand. The most severe shortfall occurs during July-October 2026, when available supply drops to 1,180,551 metric tonnes against peak requirements of 3,487,560 metric tonnes, a staggering deficit of 2,306,519 metric tonnes representing 66.1 percent of crushing demand, meaning mills will operate at only 33.9 percent of installed capacity during these critical months. Aggregating across the full annual cycle, total available cane supply of 5,000,069 metric tonnes falls dramatically short of total mill requirements of 10,462,680 metric tonnes, yielding an annual deficit of 5,461,621 metric tonnes and an average mill utilization rate of just 47.8 percent.

This means Western Kenya's sugar factories will collectively operate at less than half their installed crushing capacity, with profound economic consequences: the 5.46 million tonne deficit represents approximately 24.6 billion Kenya Shillings in foregone farmer income at current cane prices, while mills face escalating per-unit processing costs as fixed expenses for depreciation, debt service, and administrative overhead must be spread across severely reduced output volumes.

Period	Available Cane (MT)	Mill Requirement (MT)	Deficit (MT)	Deficit %	Mill Utilization
Nov-Dec 2025	1,069,936	1,743,780	673,844	38.60%	61.40%
Jan-Mar 2026	1,440,431	2,615,670	1,175,239	44.90%	55.10%
Apr-Jun 2026	1,309,151	2,615,670	1,306,519	50.00%	50.00%
Jul-Oct 2026	1,180,551	3,487,560	2,306,519	66.10%	33.90%
Annual	5,000,069	10,462,680	5,461,621	52.2%	47.8%

5.4 Feature Importance Analysis

The Random Forest model provides built-in capability to assess which predictor variables contribute most significantly to yield predictions, offering valuable insight into what factors drive sugarcane productivity variation across the Western Catchment. Drone-measured field area emerged as the single most influential predictor, accounting for 33 percent of total variable importance, which directly validates the investment in high-resolution drone mapping for precise boundary delineation, accurate area measurement fundamentally constrains yield prediction accuracy regardless of how well other variables capture productivity differences. Historical tonnes of cane per hectare derived from previous seasons' mill weighbridge records contributed 23 percent of importance, demonstrating that farm-specific production history captures persistent differences in management quality, soil fertility, and microclimate conditions that current-season spectral observations alone cannot fully reflect.

Yield threshold variables associated with crop cycle stage (Plant Cane, Ratoon 1, Ratoon 2, Ratoon 3+) contributed 20 percent, confirming that the well-documented decline in yields across successive ratoon crops represents a major source of production variation that must be explicitly accounted for in any prediction model. Among the vegetation indices derived from satellite and drone imagery, the Normalized Difference Red Edge Index (NDRE) contributed 7 percent, followed by Green Chlorophyll Index (GCI) at 6 percent, Normalized Difference Vegetation Index (NDVI) at 5 percent, Green Normalized Difference Vegetation Index (GNDVI) at 4 percent, and Wide Dynamic Range Vegetation Index (WDRVI) at 1 percent. The prominence of NDRE among spectral indices aligns with international research findings that red-edge spectral bands maintain sensitivity to chlorophyll content and canopy health even in dense mature sugarcane where conventional NDVI values saturate. Crop class categorical variables contributed the remaining 1 percent of importance.

The Feature Importance Analysis and crop cycle composition are intrinsically linked through the relationship between ratoon age and yield productivity. The Random Forest model identified Yield Threshold as the second most influential variable at 20.0% importance, which directly correlates with the documented crop cycle distribution of 31:26:26:17. This connection becomes evident when examining the elevated R3+ proportion of 17%, which significantly exceeds the industry-recommended 10% benchmark. Third ratoon and older cane exhibit progressive yield decline due to accumulated soil nutrient depletion, increased pest and disease pressure, and deteriorating stool vigor over successive harvests. The model's heavy weighting on Yield Threshold reflects the substantial yield variability introduced by this aging ratoon population—fields dominated by R3+ cane demonstrate markedly different productivity patterns compared to younger plant cane or early ratoons. Essentially, the machine learning algorithm has independently validated what agronomic science predicts: the disproportionate presence of older ratoon cycles in the Western catchment creates systematic yield depression that must be accounted for in accurate production forecasting. This alignment between algorithmic feature importance and observed crop cycle demographics strengthens the model's credibility while simultaneously highlighting a critical management intervention point—accelerating the replanting of aging R3+ fields would not only improve actual yields but would also reduce prediction uncertainty by normalizing the crop age distribution toward the 30:30:30:10 standard.

Variable	**Importance (%)**
Drone-measured Area	33%
Tonnes of Cane/Ha (Historical)	23%
Yield Threshold	20%
NDRE	7%
GCI	6%
NDVI	5%
GNDVI	4%
WDRVI	1%
Crop Class	1%

Figure 7: Google Earth Engine Display Dashboard

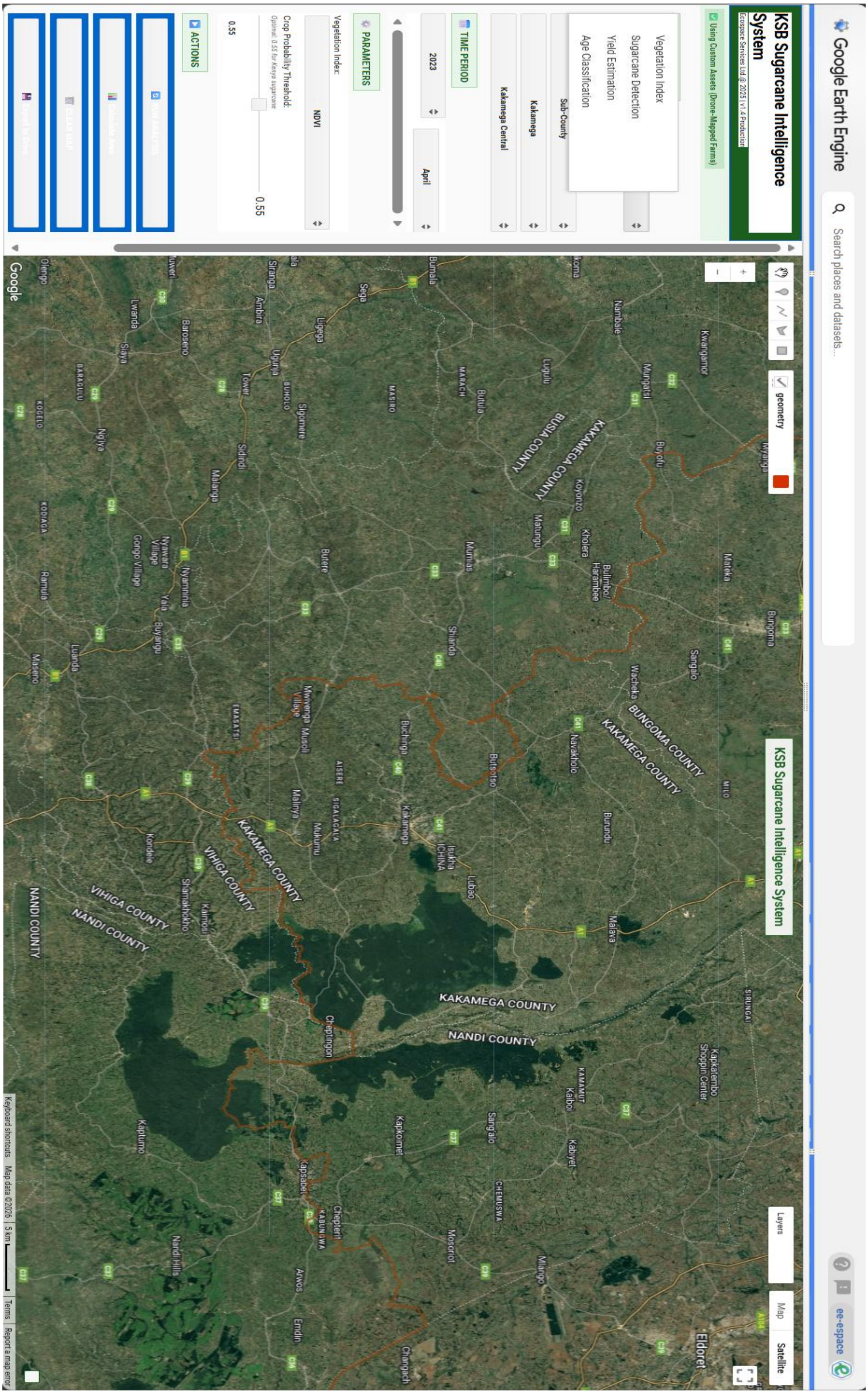
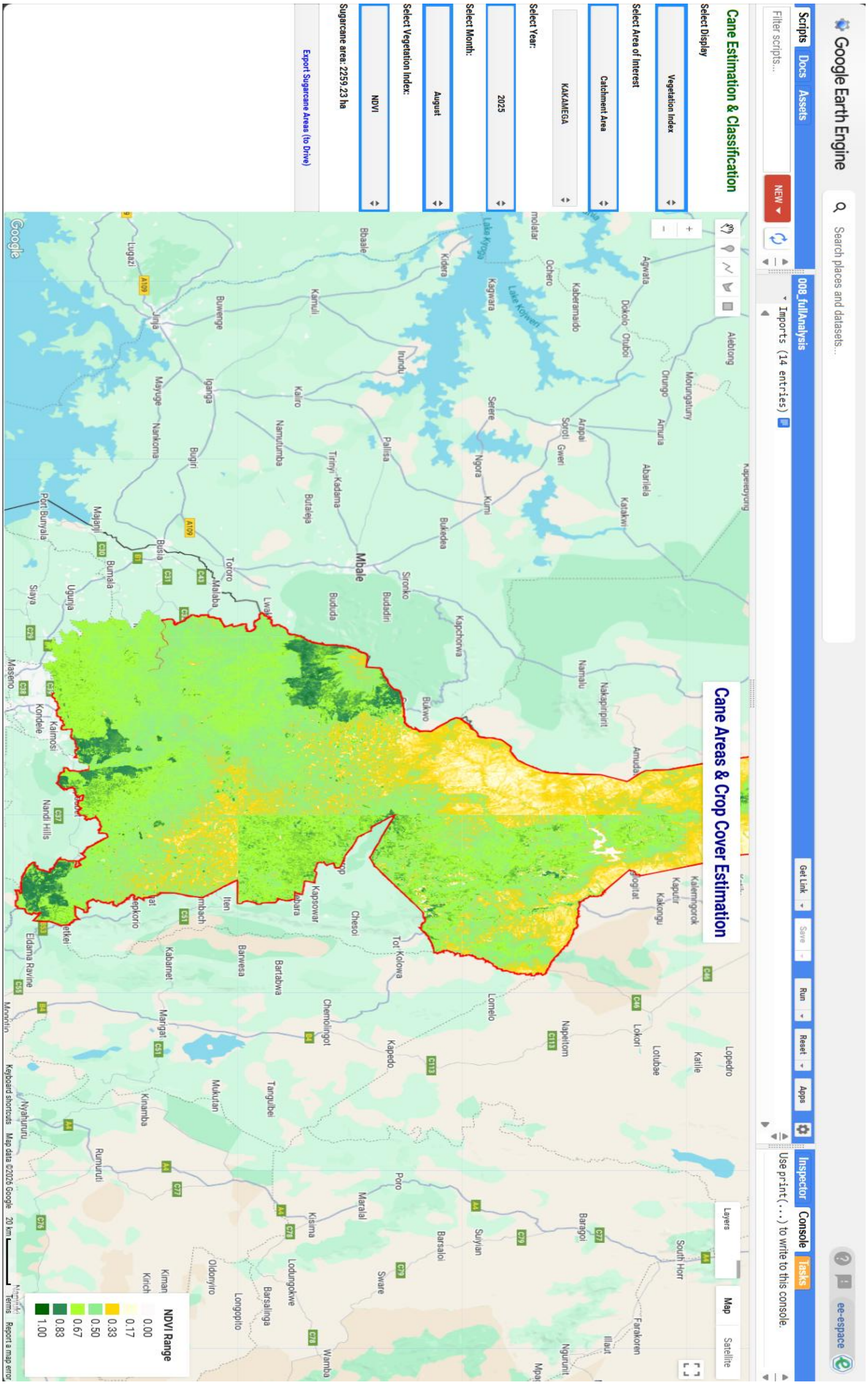


Figure 8: Earth Engine NDVI Result



5.5 Statistical Validation

The statistical validation through paired t-test analysis provides foundational evidence of model viability while simultaneously illuminating pathways for systematic improvement. The observed mean difference of 1.0019 tonnes between drone-based predictions and census-reported farmer estimates, coupled with a p-value of 0.1483 exceeding the 0.05 significance threshold, confirms that the current model achieves statistical equivalence with traditional census methodologies at aggregate catchment levels. However, this equivalence should be interpreted not as validation of model perfection but rather as confirmation that the predictive framework has established a reliable baseline from which iterative enhancements can proceed.

The variance observed at individual field levels suggests that expanding the training dataset through additional sampling campaigns across diverse agro-ecological zones, soil types, and microclimatic conditions would strengthen the model's generalization capabilities. Furthermore, implementing continuous model retraining protocols, whereby each harvest season's actual yield data feeds back into the algorithm, would enable the Random Forest model to progressively calibrate its predictions against ground-truth outcomes, reducing systematic biases that may currently exist within specific crop cycle stages or geographic clusters.

The integration of supplementary datasets, including historical weather patterns, soil nutrient mapping, irrigation infrastructure coverage, and disease incidence records, would provide the model with additional explanatory variables that could account for yield variability not currently captured by vegetation indices alone. This adaptive machine learning approach transforms the validation exercise from a static pass-fail assessment into a dynamic improvement roadmap, positioning the drone-based system for progressively enhanced accuracy with each operational cycle while maintaining its inherent advantages in spatial precision and field-level discrimination that census methodologies fundamentally cannot replicate.

6 Discussion

6.1 Interpretation of Model Performance

The R^2 value of 0.923 indicates operationally useful prediction accuracy, exceeding the commonly accepted threshold of $R^2 = 0.85$ for agricultural forecasting. The model successfully captured key factors affecting yield variation including environmental conditions, crop health, and growth patterns.

The 16.99% average overestimation in census-reported areas has significant implications:

a) Mills relying on census figures would expect substantially more cane than actually exists.
b) Resulting shortfalls disrupt crushing schedules and leave mill capacity underutilized.
c) Economic impact: 5.46 million tonne deficit represents approximately KShs 24.6 billion in foregone farmer income

6.2 Comparison with Existing Studies

Results align with international research benchmarks:

a) Australian studies (Field Crops Research): $R^2 = 0.96$
b) Sri Lankan studies: $R^2 = 0.91$
c) Global Random Forest meta-analysis: mean $R^2 \approx 0.96$

The prominence of red-edge indices (NDRE) aligns with Ethiopian Sentinel-2 studies showing red-edge bands crucial for enhancing prediction precision.

6.3 Practical Implications

6.3.1 For Mill Operations

a) Real-time visibility into cane maturity and availability
b) Optimized harvest scheduling and logistics coordination
c) Early detection of fields reaching harvest maturity

6.3.2 For Farmers

a) More accurate payments based on objective measurements
b) Transparent boundary verification
c) Access to precision agriculture tools

6.3.3 Economic Benefits

a) Current census costs: KShs 5-15 million annually
b) Drone-assisted costs: KShs 2-5 million annually (after initial investment)
c) Potential annual savings: KShs 3-10 million
d) Payback period: 4-6 months

6.4 Limitations and Challenges

a) Field Fragmentation: Average plot size of 0.48 Ha increases per-hectare mapping costs
b) Weather Constraints: Persistent cloud cover disrupts both drone and satellite data collection
c) Spectral Similarity: Sugarcane, maize, and Napier grass share similar spectral signatures
d) Equipment Limitations: Battery capacity and data storage reduced daily operational capacity
e) Power Supply: Frequent outages disrupted data processing workflows
f) Transportation: Poor road conditions consumed substantial transit time

6.5 Recommendations for Future Research

a) Additional Data Integration: Soil type, weather data, pest/disease information
b) Temporal Analysis: Time-series monitoring throughout growth cycles
c) Model Optimization: Seasonal recalibration with each harvest season's data

d) Regional Calibration: Local training data for different agro-ecological zones
e) Decision Support Integration: Real-time recommendations for farm management

7 Conclusion

7.1 Summary of Key Findings

a) Drone-assisted mapping significantly improves accuracy: 16.99% overestimation documented in conventional census methods eliminated through objective drone measurement
b) Machine learning achieves high prediction accuracy: $R^2 = 0.923$ matches global best practice standards for sugarcane yield estimation.
c) Crop classification challenges addressable: 89.7% overall accuracy achieved through multi-temporal analysis and phenological profiling.
d) KSB staff capacity established: Five officers trained in GIS, Remote Sensing, and Machine Learning applications.
e) Infrastructure investment justified: KShs 8.7 million total investment recoverable within 4-6 months through operational savings

7.2 Implications for Sugarcane Farming in Kenya

a) Industry Impact: Supports 8 million Kenyans; 309,017 registered farmers across 149,438 hectares
b) Supply Crisis: 52.2% annual deficit (5.46 million MT) against mill requirements
c) Modernization Potential: Drone technology viable for routine cane availability surveys
d) Scalability: Applicable from smallholder (0.48 Ha average) to large-scale plantations

7.3 Limitations of the Research

a) Model trained on specific regions; generalization requires regional calibration
b) Temporal snapshots may not capture rapid condition changes
c) Environmental factors (soil, irrigation, pests) not fully incorporated
d) Current equipment inventory limits simultaneous multi-region deployment

7.4 Recommendations for Future Implementation

7.4.1 Immediate (0-6 months)

a) Procure essential drone accessories (KShs 2.035 million)
b) Complete Western Catchment digitization
c) Train five additional KSB officers

7.4.2 Medium-term (6-18 months)

a) Acquire high-resolution satellite imagery for all catchments
b) Deploy regional drone hubs (Western, Nyanza, Coastal)
c) Integrate with mill scheduling systems

7.4.3 Long-term (18-36 months)

a) Achieve full national coverage
b) Implement real-time monitoring dashboard
c) Establish data sharing agreements with stakeholders

7.5 Final Thoughts

The collaboration between Kenya Sugar Board and Ecospace Services Ltd. has demonstrated the transformative potential of drone and satellite technology for modernizing agricultural monitoring in Kenya. With proper implementation, Kenya Sugar Board will be positioned as a regional leader in data-driven agricultural management, demonstrating innovation that peer institutions across Africa may seek to emulate.

The foundation has been established; the path forward is clear; the opportunity for transformative impact awaits execution.